STAT293: Tutorial 2 Solutions

1. (a) $E(Y_{ij}) = E(\mu) + E(\alpha_i) + E(\epsilon_{ij}) = \mu + \alpha_i$
   (b) $Var(Y_{ij}) = Var(\mu) + Var(\alpha_i) + Var(\epsilon_{ij}) = \sigma_i^2$

2.

Vehicle Type	Sample Size, n_i	Mean, $\bar{y}_{i.}$	Standard Deviation, s_i
Small car	45	4.59	0.70
Large car	102	4.88	0.64
Truck or bus	18	6.24	0.90

(a) Using:

$$SSE = \sum_{i=1}^{k}(n_i - 1)s_i^2, \qquad SST = \sum_{i=1}^{k}(n_i \bar{Y_{i.}}^2) - \frac{(\sum n_i \bar{Y_{i.}})^2}{\sum n_i}$$

```
#SSE = sum{(ni-1)*si^2}
SSE<-44*0.70^2 + 101*0.64^2 + 17*0.90^2
SSE
```

```
## [1] 76.6996
```

```
#first term in SST
sst1<-45*4.59^2 + 102*4.88^2 + 18*6.24^2
#second term in SST
sst2<-((45*4.59 + 102*4.88 + 18*6.24)^2)/(45+102+18)

SST<-sst1-sst2
SST
```

```
## [1] 36.28551
```

```
MSE<-SSE/(45+102+18 -3)
MSE
```

```
## [1] 0.4734543
```

```
MST<-SST/(3-1)
MST
```

```
## [1] 18.14276
```

```
Fval<-MST/MSE
Fval
```

```
## [1] 38.31997
```

```
pval<-1-pf(Fval, 2,162)
pval
```

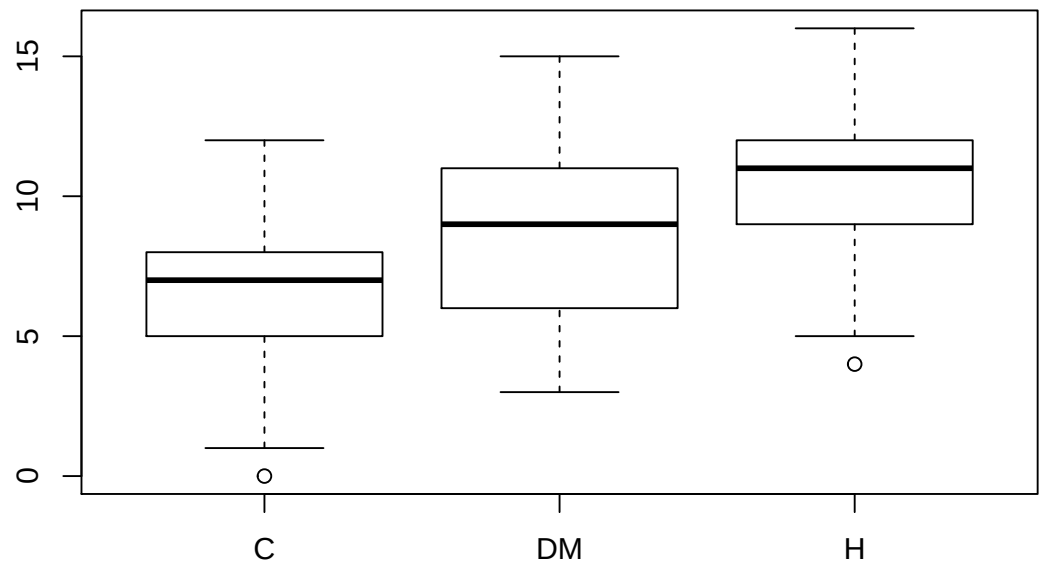
```
## [1] 2.364775e-14
```

(b) $H_0 : \mu_1 = \mu_2 = \mu_3$ vs H_1 : not all means are equal. We get $F = 38.32$ with 2 and 162 degrees of freedom, and p-value=2.36×10^{-14}. There is very strong evidence that mean maneuver times differ among vehicle types.
A separate left-turn phase at the traffic light might be of benefit for trucks and buses to allow them time to complete the left-hand turn. This is because their mean maneuver time is higher than that for other vehicle types.

3. (a) $W = \frac{\sum_{j=1}^{2}(Y_j - \bar{Y})^2}{\sigma^2} \sim \chi^2_1$
   (b) We know $\sum_i W_i \sim \chi^2_{\sum n_i}$ if $W_i \sim \chi^2_{n_i}$. Therefore $X_1 = \sum_{i=1}^{3} W_i \sim \chi^2_{10}$
   (c) $\frac{W_1/n_1}{W_2/n_2} \sim F_{n_1,n_2}$ if $W_1 \sim \chi^2_{n_1}$ and $W_2 \sim \chi^2_{n_2}$. Therefore $X_2 = \frac{W_1/10}{W_2/6} \sim F_{10,6}$.

4. (a)

```
honey<-read.csv("crdhoneycough.csv", header=T)
boxplot(score ~ treatment, data=honey)
```




```
tapply(honey$score, honey$treatment,mean)
```

```
##         C        DM         H
##  6.513514  8.333333 10.714286
```

```
tapply(honey$score, honey$treatment,sd)
```

```
##        C       DM        H
## 2.940348 3.256404 2.855041
```

The boxplot indicates that mean changes in symptom scores may differ among groups, with honey leading to greater improvement compared to the other treatments.

(b)

```r
fit1<-lm(score~treatment, data=honey)
summary(fit1)
```

```
##
## Call:
## lm(formula = score ~ treatment, data = honey)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -6.7143 -1.7143  0.4865  1.6667  6.6667
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   6.5135     0.4958  13.137  < 2e-16 ***
## treatmentDM   1.8198     0.7221   2.520   0.0133 *
## treatmentH    4.2008     0.7111   5.907 4.62e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.016 on 102 degrees of freedom
## Multiple R-squared:  0.2556, Adjusted R-squared:  0.241
## F-statistic: 17.51 on 2 and 102 DF,  p-value: 2.902e-07
```

$\hat{\mu}_H - \hat{\mu}_C = 4.2008$ indicating that mean score change for honey is higher (based on positive value) than that for the control group by 4.2008.

(c) $\hat{\mu}_C = 6.5135$ $\hat{\mu}_{DM} = \hat{\mu}_C + (\hat{\mu}_{DM} - \hat{\mu}_C) = 6.5135 + 1.8198 =$ `8.33`. $\hat{\mu}_H = \hat{\mu}_C + (\hat{\mu}_H - \hat{\mu}_C) = 6.5135 + 4.2008 =$`10.71`.

(d)

```r
anova(fit1)
```

```
## Analysis of Variance Table
##
## Response: score
##            Df Sum Sq Mean Sq F value    Pr(>F)
## treatment   2 318.51 159.255   17.51 2.902e-07 ***
## Residuals 102 927.72   9.095
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

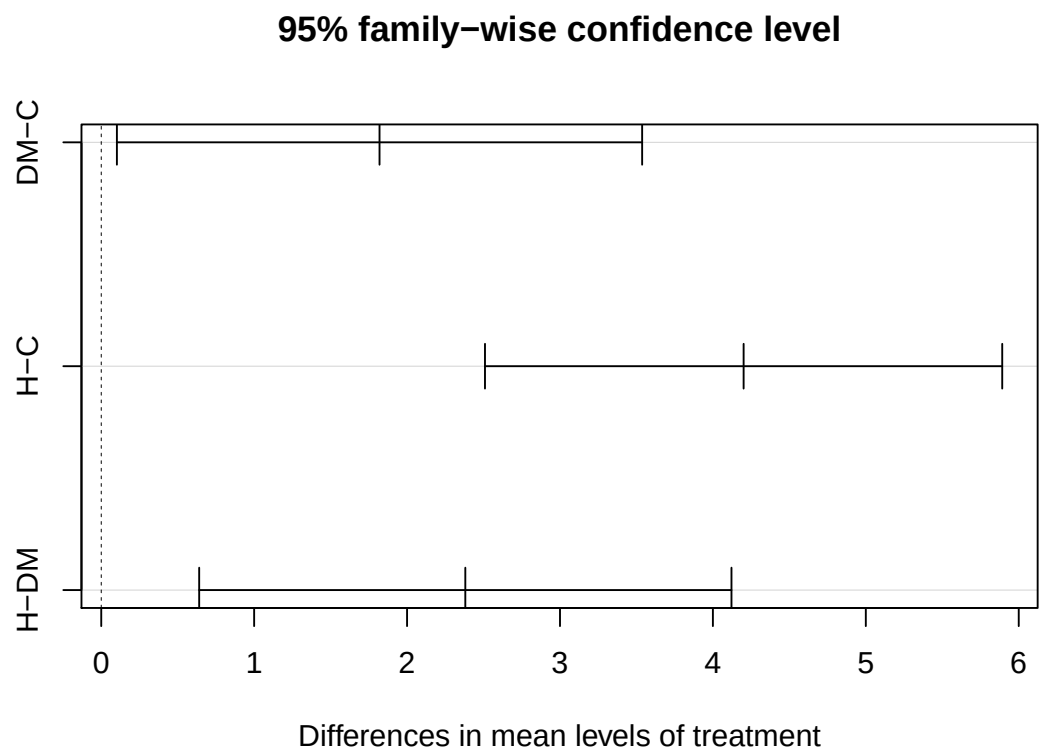
$H_0 : \mu_C = \mu_{DM} = \mu_H$ vs H_1 : not all means are equal. $F = 17.51$ with 2 and 102 degrees of freedom, and $p - value = 2.902 \times 10^{-7}$. There is very strong evidence that mean scores differ among groups.

(e)

```r
TukeyHSD(aov(score~treatment,data=honey))
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = score ~ treatment, data = honey)
##
## $treatment
##          diff       lwr      upr     p adj
## DM-C 1.819820 0.1023625 3.537277 0.0351562
## H-C  4.200772 2.5094509 5.892094 0.0000001
## H-DM 2.380952 0.6405157 4.121389 0.0043728
```

```r
plot(TukeyHSD(aov(score~treatment,data=honey)))
```



There is sufficient evidence that all treatment pairs differ in mean scores. The greatest difference is between honey and no treatment.

(f) Based on these results, buying the over-the-counter cough remedy would seem to be a waste of money! The parent would be better off using the honey they have at home as it resulted in higher symptom improvement scores compared to the over-the-counter cough remedy.